A key component of the Paris Olympic drone mobility plan is the deployment of electric vertical takeoff and landing (eVTOL) aircraft from companies such as Volocopter, which are being tested for short-distance passenger transit across selected city zones. Specially designated vertiports compact landing and take-off platforms integrated into strategic locations – are being developed to support these flights. These vertiports are linked with Paris's broader transportation networks, demonstrating how aerial mobility systems may eventually integrate seamlessly with metro, bus, and rail services to create a multi-layered, hybrid transportation ecosystem.

Beyond passenger movement, UAVs will also support logistical and surveillance functions during the games. High-resolution imaging drones will provide real-time aerial monitoring of crowd movement, traffic flow, and venue security conditions. Medical logistics drones may be used to rapidly move defibrillators, blood units, and emergency equipment between event zones and medical stations, particularly when ground pathways are congested. These applications underscore the flexibility of UAVs in managing large events where timely and coordinated response systems are critical (KA AL-Dosari, et al. 2025).

While the Paris Olympic initiative demonstrates strong innovation potential, it also reveals several challenges requiring careful management. These include airspace safety coordination, noise considerations, public trust, regulatory harmonization with European Union aviation laws, and cybersecurity protection for autonomous vehicles. Nevertheless, Paris's proactive deployment showcases how international events can serve as catalysts for accelerated UAM testing and adoption, laying the foundation for long-term implementation of drone mobility in major world cities.

8.10.4 Case Study: United States – UAV-based infrastructure